

Carbon Pricing and Co-Abatement of Toxic Emissions

Evidence from Alberta's SGER/TIER Regulation

Oluwafemi Ojo

May 2026

Swipe →

Outline

01 The Question

02 US Benchmark

03 Alberta's Policy

04 Data

05 Empirical Strategy

06 Results

07 Event Study

08 Takeaway

01 The Question

Carbon prices are designed to cut CO₂ — but do they also **reduce toxic co-pollutants**?

Mechanism: A carbon price raises the cost of fossil-fuel combustion. Firms respond by cutting output, switching fuels, or adopting cleaner technology. Each channel also reduces *co-emitted* toxic substances — mercury, arsenic, lead, VOCs.

US evidence (cap-and-trade)

Pham & Roach (2024): RGGI cut coal-utility toxic releases by **62%**

Canadian question (intensity standard)

Does Alberta's SGER/TIER produce **comparable co-abatement**?

02 US Benchmark: Pham & Roach (2024)

Setting RGGI cap-and-trade, 11 NE states

Data US Toxics Release Inventory (TRI)

Sample Coal electric utilities, 2000–2019

N 4,831 facility-years, 243 facilities

Design: Two-period DiD

- Facility, state, and year FEs
- SE clustered by facility
- Reference: non-RGGI states

Results

Phase 1 (2009)	−45.4%
Phase 2 (2014)	−30.8%
Combined	−62.2%

Mechanism: hard cap forces output cuts → less coal ash, fewer stack emissions

03 Alberta's Carbon Policy

Period	Regulation	Rate (CAD/t)
2007–2017	SGER	\$15–30
2018–2019	CCIR	\$20–30
2020–2022	TIER	\$30–50

Coverage: facilities $\geq 100,000$ t CO₂e/yr

RGGI vs. SGER/TIER

RGGI: hard cap on total tonnes — quantity instrument

SGER/TIER: intensity standard — firms can grow output if intensity falls

Key: intensity standard weakens production-cut incentive → expect smaller co-abatement

04 Data

Source National Pollutant Release
Inventory
(NPRI), ECCC, 2000–2022

Industry filter 5 energy NAICS codes:
211110/211113/211114 — oil & gas extraction
221112 — fossil-fuel power generation
324110 — petroleum refineries

Excluded British Columbia (carbon tax
2008
contaminates control group)

Sample

Observations	64,316
Facilities	12,710
AB (treated)	9,895
Control prov	2,815
Years	2000–2022

Outcomes:

$$lsumemission = \log(1 + R_{it}^{air})$$

$$lsumreleases = \log(1 + R_{it}^{all})$$

Two-way fixed effects DiD:

$$y_{it} = \underbrace{\beta_1(Alberta_i \times SGER_t)}_{\text{Phase 1 effect}} + \underbrace{\beta_2(Alberta_i \times TIER_t)}_{\text{Phase 2 effect}} + \mathbf{X}'_{it}\boldsymbol{\gamma} + \alpha_i + \delta_t + \varepsilon_{it}$$

$Alberta_i = 1$ if facility in Alberta

$SGER_t = 1[t \geq 2007]$

$TIER_t = 1[t \geq 2020]$

$\mathbf{X}_{it}$: log GDP, fossil-fuel share, coal/gas price

α_i : facility FE δ_t : year FE

SE clustered by facility

Parallel trends test

Event study (ref = 2006)

Pre-trend joint Wald test:

$F(3) = 1.70, p = 0.166$

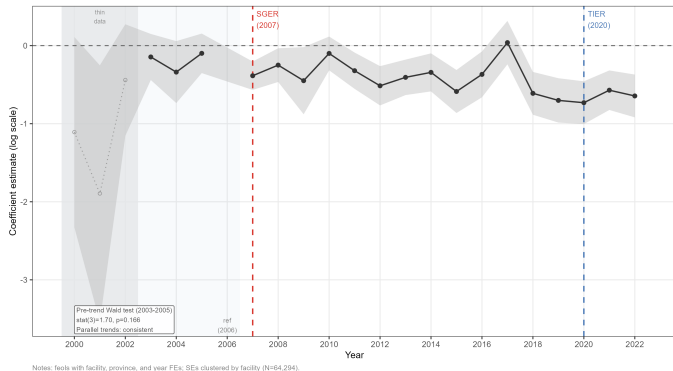
✓ **Fail to reject — consistent**

06 Results: US vs. Canada

	<i>Log Air Emissions</i>		<i>Log Total Releases</i>	
	<i>US (RGGI)</i>	<i>Canada (SGER/TIER)</i>	<i>US (RGGI)</i>	<i>Canada (SGER/TIER)</i>
$\hat{\beta}_1$ (Phase 1)	-0.605** (0.280)	-0.193*** (0.064)	-0.870*** (0.308)	-0.194*** (0.063)
$\hat{\beta}_2$ (Phase 2)	-0.369 (0.308)	-0.308*** (0.068)	-0.949*** (0.398)	-0.306*** (0.068)
<i>Begin %</i>	-45.4%	-17.5%	-56.7%	-17.6%
<i>Lower %</i>	-30.8%	-26.5%	-61.7%	-26.4%
Total %	-62.2%	-39.4%	-83.4%	-39.4%

SE clustered by facility. Bootstrap 95% CI for $\hat{\beta}_1$: $[-0.31, -0.06]$; for $\hat{\beta}_2$: $[-0.42, -0.18]$. ** $p < 0.05$,

*** $p < 0.01$.



2003–2005: flat near zero

Pre-trends hold ✓

2007: immediate drop at SGER introduction

2020: further decline at TIER introduction

2022: -0.64 log pts ($\approx -47\%$)

Open circles (2000–2002): < 120 AB obs

Canadian oil & gas sector:

99.2% of NPRI releases
are to **air**

Mean air: 384,000 kg/facility-yr

Mean non-air: 1,655 kg/facility-yr

$\text{cor}(l_{\text{summission}}, l_{\text{sumreleases}}) =$
0.999

US coal utilities:

Coal ash $\rightarrow$ land disposal

Scrubber slurry $\rightarrow$ water

So $l_{\text{sumreleases}} \gg l_{\text{summission}}$

Implication:

In Canada, the mechanism is
output curtailment, not
end-of-pipe or media substitution.

Fewer barrels/m³ produced

$\rightarrow$ less venting/flaring

$\rightarrow$ lower toxic air emissions

▶ **Co-abatement exists under intensity pricing.**

SGER still reduced toxic air emissions by **17.5%** — sufficient to shift firm technology.

▶ **Tighter prices deliver larger gains.**

TIER's higher price added a further **26.5%**. Price level matters, not just instrument type.

▶ **But cap-and-trade delivers more.**

Canada's **-39%** is half the US RGGI **-62%** benchmark. A hard cap removes the output-expansion escape valve.

As Canada's carbon price escalates toward \$170/tCO₂e by 2030, further co-abatement can be expected — these health co-benefits should enter policy benefit-cost analyses.

References

Pham, H. & Roach, T. (2024). Spillover benefits of carbon dioxide cap and trade: Evidence from the Toxics Release Inventory. *Economic Inquiry*, 62(1).

Callaway, B. & Sant'Anna, P.H.C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230.

ECCC (2024). National Pollutant Release Inventory — bulk data files.
`open.canada.ca`

Statistics Canada (2024). Tables 36-10-0222-01, 25-10-0029-01, 18-10-0268-01, 18-10-0004-01.

Full replication code:

`github.com/oluwafemiojo/`

`Carbon-Pricing-and-Co-Abatement-of-Toxic-Emissions-Evidence-from-Alberta`